

⚡ OHM'S LAW | ओम का नियम ⚡

◆ Statement (English):

✦ At constant temperature, current (I) is directly proportional to voltage (V).

◆ कथन (Hindi):

✦ समान तापमान पर धारा (I), वोल्टेज (V) के समानुपाती होती है।

$$\rightarrow V \propto I$$

📌 Formula | सूत्र

$$\text{⚡ } V = I \times R$$

$$\text{💡 } I = V / R$$

$$\text{🧱 } R = V / I$$

🧠 Meaning | अर्थ

🔌 Voltage (V) → Electrical pressure | विद्युत दाब

💡 Current (I) → Flow of charge | आवेश का प्रवाह

🧱 Resistance (R) → Opposition to current | धारा का विरोध

✦ Important Points | महत्वपूर्ण बिंदु

- Unit of V → Volt (V) ⚡ | वोल्ट
- Unit of I → Ampere (A) 💡 | एम्पियर
- Unit of R → Ohm (Ω) 🧱 | ओम
- Valid for DC circuits
- Temperature constant hona chahiye 🔧

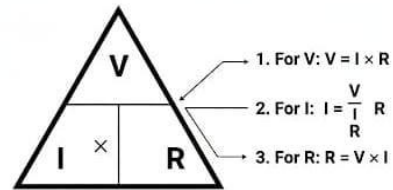
✦ Important Points

- Ohm's law DC circuits ke liye apply hota hai 💡
- Temperature constant hona chahiye 🔧
- Resistance constant ho to $V \propto I$
- Non-ohmic devices (diode, filament lamp) par apply nahi hota ✖
- Circuit analysis ka base law hai ⚡

💡 Memory Trick

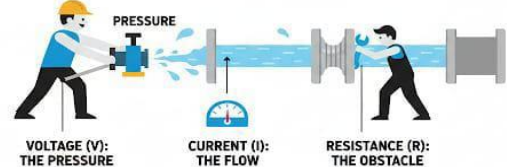
👉 Voltage badhao → Current badhega (R constant ho)

OHM'S LAW TRIANGLE



Example: $V = 12V$, $R = 4\Omega$, $I = ?$
 $I = 12 / 4 = 3A$

OHM'S LAW: WATER PIPE ANALOGY



↑ INCREASE V → INCREASE I
↓ INCREASE R: DECREASE I

$$I = \frac{V}{R} R$$

